

**THE UNIVERSITY OF HONG KONG  
FACULTY OF BUSINESS AND ECONOMICS**

**MSBA7015 Service Operations Management**

| <b>GENERAL INFORMATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                            |
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| <b>Instructor:</b> Dr. Huiyin Ouyang<br><b>Email:</b> <a href="mailto:oyhy@hku.hk">oyhy@hku.hk</a><br><b>Office:</b> Room 1321, K.K.Leung Building<br><b>Consultation times:</b> By appointment<br><b>Module / Class:</b> Module 5<br><b>Class meetings:</b> Wednesday and Saturday, 18:30-21:30<br><b>Pre-requisites:</b> MSBA7003 and MSBA7004<br><b>Mutually exclusive:</b> N/A<br><b>Course Website:</b> Moodle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                            |
| <b>COURSE DESCRIPTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                            |
| Services run the modern economy — healthcare, airlines, e-commerce, ridesharing, and AI-powered platforms all live or die by how well they deliver their services. This course gives you the analytical edge to design and optimize service operations in a data-driven world. This course equips students to design, measure, improve, price, staff, predict, and scale service systems. Across ten lectures, students use service blueprints and customer-variability frameworks; SERVQUAL, the gaps model, and statistical process control; lean process analysis and bottleneck management; revenue-management and queueing models; predictive analytics for churn and no-shows; and platform and AI-native service design. Through real cases and hands-on analytics, you'll build skills that are directly valued in consulting, tech, healthcare, finance, and platform businesses wherever operations meets data. |                                            |
| <b>COURSE OBJECTIVES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                            |
| <ol style="list-style-type: none"> <li>1. Diagnose service systems using service blueprints, service packages, customer variability, and service-quality measures.</li> <li>2. Quantify flow, capacity, waiting, and revenue using value-stream mapping, takt time, bottleneck analysis, queueing models, overbooking, and differentiated pricing.</li> <li>3. Analyze churn, survival, no-shows, and customer segments, and evaluate platform and AI-native service designs.</li> <li>4. Integrate multiple analytical lenses, sequence interventions, and communicate evidence-based recommendations to managerial audiences.</li> </ol>                                                                                                                                                                                                                                                                                |                                            |
| <b>Programme Learning Outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                            |
| PLO1: Acquisition and internalization of knowledge of the programme discipline<br>PLO2: Application and integration of knowledge<br>PLO3: Inculcating professionalism and leadership<br>PLO4: Developing global outlook<br>PLO5: Mastering communication skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |
| <b>COURSE LEARNING OUTCOMES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |
| <b>Course Learning Outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Aligned Programme Learning Outcomes</b> |
| CLO1 Students will diagnose service systems using service-design, customer-variability, and quality-measurement frameworks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CLO1 - PLO1                                |
| CLO2 Students will apply quantitative tools for process flow, revenue, capacity, and delay management.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CLO2 - PLO1, 2                             |
| CLO3 Students will analyze customer behaviour and demand data and evaluate platform and AI-enabled service designs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CLO3 - PLO2                                |
| CLO4 Students will integrate analytical results and communicate evidence-based                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CLO4 - PLO2, 3, 5                          |

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|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------|
| service-improvement recommendations.               |                                                                                                                                                                                                                                                                                          |                                |                                         |
| <b>COURSE TEACHING AND LEARNING ACTIVITIES</b>     |                                                                                                                                                                                                                                                                                          |                                |                                         |
| <b>Course Teaching and Learning Activities</b>     |                                                                                                                                                                                                                                                                                          | <b>Expected learning hours</b> | <b>Study Load (% of study)</b>          |
| T&L1. Interactive lectures, cases, and discussions |                                                                                                                                                                                                                                                                                          | 30                             | 25%                                     |
| T&L2. Skill-building exercises and preparation     |                                                                                                                                                                                                                                                                                          | 30                             | 25%                                     |
| T&L3. Individual assignments and practice          |                                                                                                                                                                                                                                                                                          | 30                             | 25%                                     |
| T&L4. Group project                                |                                                                                                                                                                                                                                                                                          | 30                             | 25%                                     |
| Total                                              |                                                                                                                                                                                                                                                                                          | 120                            | 100%                                    |
|                                                    |                                                                                                                                                                                                                                                                                          |                                |                                         |
| <b>Assessment Methods</b>                          | <b>Brief Description (Optional)</b>                                                                                                                                                                                                                                                      | <b>Weight</b>                  | <b>Aligned Course Learning Outcomes</b> |
| A1. Individual assignments                         | Assignments are designed to give you practice of the concepts being covered in class.                                                                                                                                                                                                    | 40%                            | CLO1, 2, 3, 4                           |
| A2. Participation                                  | Students should read all required readings before class and prepare to discuss the contents. Contribution is judged based on the extent to which you appear prepared, and on the relevance and depth of your comments/questions. Both in class and online contributions will be counted. | 20%                            | CLO1, 2, 4                              |
| A3. Group project                                  | 4-5 students in each group. Students apply the knowledge learnt in this course in one practical situation (either designing a new service process or improving an existing service process).                                                                                             | 40%                            | CLO1, 2, 3, 4                           |
|                                                    | Total                                                                                                                                                                                                                                                                                    | 100%                           |                                         |
| <b>STANDARDS FOR ASSESSMENT</b>                    |                                                                                                                                                                                                                                                                                          |                                |                                         |
| <b>Course Grade Descriptors</b>                    |                                                                                                                                                                                                                                                                                          |                                |                                         |
| A+, A, A-                                          | <ul style="list-style-type: none"> <li>• Demonstrates comprehensive and accurate understanding of relevant concepts.</li> <li>• Applies analytical tools with excellent judgment and originality.</li> <li>• Communicates recommendations clearly and professionally.</li> </ul>         |                                |                                         |
| B+, B, B-                                          | <ul style="list-style-type: none"> <li>• Demonstrates good understanding of relevant concepts.</li> <li>• Applies analytical tools accurately with sound judgment.</li> <li>• Communicates recommendations effectively.</li> </ul>                                                       |                                |                                         |
| C+, C, C-                                          | <ul style="list-style-type: none"> <li>• Demonstrates adequate understanding of the main concepts.</li> <li>• Applies appropriate tools with some errors or limited interpretation.</li> <li>• Communicates an acceptable recommendation.</li> </ul>                                     |                                |                                         |
| D+, D                                              | <ul style="list-style-type: none"> <li>• Demonstrates the minimum acceptable understanding.</li> <li>• Applies tools incompletely or with material errors.</li> <li>• Communicates a marginally acceptable recommendation.</li> </ul>                                                    |                                |                                         |
| F                                                  | <ul style="list-style-type: none"> <li>• Demonstrates insufficient understanding of the concepts.</li> <li>• Contains major analytical or computational errors.</li> <li>• Does not communicate a defensible recommendation.</li> </ul>                                                  |                                |                                         |

| <b>Assessment Rubrics for Each Assessment</b> |                                                                                                                                                                                                                                                |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Written assignments and group project</b>  |                                                                                                                                                                                                                                                |
| A+, A, A-                                     | <ul style="list-style-type: none"> <li>• Strong command of relevant concepts and methods.</li> <li>• Accurate computation and well-supported interpretation.</li> <li>• Original, practical recommendations presented clearly.</li> </ul>      |
| B+, B, B-                                     | <ul style="list-style-type: none"> <li>• Good command of relevant concepts and methods.</li> <li>• Mostly accurate computation and sound interpretation.</li> <li>• Useful recommendations presented effectively.</li> </ul>                   |
| C+, C, C-                                     | <ul style="list-style-type: none"> <li>• Basic command of the concepts and methods.</li> <li>• Reasonably accurate work with some analytical gaps.</li> <li>• Acceptable recommendations and presentation.</li> </ul>                          |
| D+, D                                         | <ul style="list-style-type: none"> <li>• Minimum acceptable command of the concepts.</li> <li>• Material errors or incomplete interpretation.</li> <li>• Marginally acceptable recommendations and presentation.</li> </ul>                    |
| F                                             | <ul style="list-style-type: none"> <li>• Poor command of the concepts and methods.</li> <li>• Major computational or analytical errors.</li> <li>• Unsupported recommendations or ineffective presentation.</li> </ul>                         |
| <b>Class and online participation</b>         |                                                                                                                                                                                                                                                |
| A+, A, A-                                     | <ul style="list-style-type: none"> <li>• Consistently prepared and highly engaged.</li> <li>• Contributions are relevant, original, and build on others' ideas.</li> <li>• Listens respectfully and supports productive discussion.</li> </ul> |
| B+, B, B-                                     | <ul style="list-style-type: none"> <li>• Prepared and regularly engaged.</li> <li>• Contributions are relevant and demonstrate good understanding.</li> <li>• Listens respectfully and follows course norms.</li> </ul>                        |
| C+, C, C-                                     | <ul style="list-style-type: none"> <li>• Generally prepared with occasional contributions.</li> <li>• Comments demonstrate basic understanding.</li> <li>• Participates respectfully.</li> </ul>                                               |
| D+, D                                         | <ul style="list-style-type: none"> <li>• Limited preparation or participation.</li> <li>• Contributions show minimum understanding.</li> <li>• Meets basic conduct expectations.</li> </ul>                                                    |
| F                                             | <ul style="list-style-type: none"> <li>• Unprepared or unwilling to participate.</li> <li>• Contributions do not demonstrate adequate understanding.</li> <li>• Does not meet course conduct expectations.</li> </ul>                          |

| <b>COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| See attached                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>REQUIRED/RECOMMENDED READINGS &amp; ONLINE MATERIALS</b> (e.g. journals, textbooks, website addresses etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>Recommended text: Bordoloi, Fitzsimmons and Fitzsimmons, Service Management: Operations, Strategy, Information Technology, 10th ed., McGraw Hill. Assigned chapters, lecture decks, cases, formula sheets, and assignment materials will be posted on Moodle. Students should have access to spreadsheet software; SimPy and AutoML demonstrations are optional.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>X Conducting mid-term survey in addition to SETL around the end of the semester</p> <ul style="list-style-type: none"> <li>○ Online response via Moodle site</li> <li>○ Others: _____ (please specify)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>COURSE POLICY</b> (e.g. plagiarism, academic honesty, attendance, etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>An orderly learning environment is extremely important for this course. Disruptive behaviors are inconsiderate to other students as well as to the instructor, and are absolutely unacceptable. Talking during lectures, arriving to class late, and any other disruptions of mobile devices are not allowed; students who are responsible for any of these actions will be subject to academic penalty and will be asked to leave the classroom.</b></p> <p><b>Any dishonesty - such as cheating, false representation, plagiarism, etc.—that comes to my attention will result in an F in the course.</b></p> <p><b>Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Scholastic dishonesty also includes, but is not limited to, providing false or misleading information to receive a postponement or an extension on an exam or other assignment. The responsibilities of both students and faculty with regard to scholastic dishonesty are described in detail in the Disciplinary Committee Regulations (<a href="http://www.hku.hk/pubunit/cal99/104f.htm">http://www.hku.hk/pubunit/cal99/104f.htm</a>). By teaching this course, I have agreed to observe all of the faculty responsibilities described in that document. By enrolling in this class, you have agreed to observe all of the student responsibilities described in that document. If the application of that policy statement to this class and its assignments is unclear in any way, it is your responsibility to ask me for clarification.</b></p> <p><b>Generative AI may be used for brainstorming, code debugging, and language polishing unless an assessment states otherwise. Students remain responsible for all reasoning, calculations, sources, and final writing; significant AI use must be declared, and students must be able to explain and defend submitted work.</b></p> |
| <b>ADDITIONAL COURSE INFORMATION</b> (e.g. e-learning platforms & materials, penalty for late assignments, etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>All submissions are made through Moodle by 23:59 HKT. The late penalty is 10% of the component mark per calendar day for up to three days; later submissions are not accepted unless formally approved. There is no final exam and no in-class group presentation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Tentative Lecture Plan and Key Dates**

| Session | Date       | Topic                                                  | Key coursework         |
|---------|------------|--------------------------------------------------------|------------------------|
| L1      | Sat 20 Mar | Service Foundations: Nature, Strategy and Blueprinting |                        |
| L2      | Wed 24 Mar | Customer Variability and the Service Encounter         |                        |
| L3      | Wed 31 Mar | Service Quality: SERVQUAL and the Gaps Model           | A1 released            |
| L4      | Sat 3 Apr  | Lean Service: VSM, Waste, Bottlenecks and Takt         |                        |
| L5      | Wed 7 Apr  | Revenue Management, Overbooking and Pricing            | A1 due; A2 released    |
| L6      | Sat 10 Apr | Queueing Fundamentals: M/M/1, M/M/c and Erlang C       |                        |
| L7      | Wed 14 Apr | Priority Queues, M/G/1 and Predictive Staffing         | A2 due; A3 released    |
| L8      | Sat 17 Apr | Predictive Analytics: Churn, Survival and No-Shows     |                        |
| L9      | Wed 21 Apr | Platforms, Network Effects and AI-Native Services      |                        |
| L10     | Sat 24 Apr | Integrative Diagnosis: GreenPulse Telehealth           | A3 due                 |
| Project | Mon 3 May  | Recorded group presentation                            | Project due, 23:59 HKT |